Final Project Report MATH 563

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Abstract

give some background

1 Introduction

1.1 Notation

Throughout this paper, we will denote the *proximal operator* of f with parameter λ by

$$\mathbf{prox}_{\lambda f}(v) := \arg\min_{x} \left\{ f(x) + \frac{1}{2\lambda} \|x - v\|_2^2 \right\} \tag{1}$$

The *indicator function* for a convex set C will be defined as

$$\delta_C(x) := \begin{cases} 0, & \text{if } x \in C \\ +\infty, & \text{if } x \notin C \end{cases} \tag{2}$$

We also denote the *Euclidean Projection* on a set C by

$$\mathbf{P}_C(v) := \arg\min_{x \in C} \|x - v\|_2 \tag{3}$$

1.2 Preleminaries

We first note an important relation between the proximal and subdifferential operators.

Proposition 1. Given a function $f \in \Gamma$, we have

$$\mathbf{prox}_{\lambda f} = (I + \lambda \partial f)^{-1}, \tag{4}$$

where $(I + \lambda \partial f)^{-1}$, for $\lambda > 0$, is called the **resolvent** of the operator ∂f. In particular, the proximal operator is the resolvent of the subdifferential operator [3].

Definition 1 (Maximal Monotone operator)**.** A monotone operator is **maximal** if its graph is not properly contained in the graph of any other monotone operator. If A is monotone (resp. maximal monotone) then so are A^{-1} and λA if $\lambda > 0$.

The next proposition, given in [2], will be useful in the construction of the optimizing algorithms.

Proposition 2. If F is maximal monotone, then $(I + \lambda F)^{-1}(\hat{x})$ exists and is unique for all $\hat{x} \in R^n$. The value $x = (I + \lambda F)^{-1}(\hat{x})$ of the resolvent is the unique solution of the monotone inclusion

$$\hat{x} \in x + \lambda F(x) \tag{5}$$

Preprint. Under review.

Proposition 3 (Proximal operator splitting)**.** If f is separable across two variables, so $f(x,y) = \varphi(x) + \psi(y)$, then

$$\mathbf{prox}_f(x,y) = (\mathbf{prox}_\varphi(x), \mathbf{prox}_\psi(y)) \tag{6}$$

Thus, evaluating the proximal operator of a separable function reduces to evaluating the proximal operators for each of the separable parts, which can be done independently [3].

Comment (remove after): do we need the following proposition?

Proposition 4. (Proximal operator of norms) If $f = \|\cdot\|$ is a general norm, then f^* is the indicator function on the unit ball for the dual norm $\|\cdot\|_*$, namely

$$\|\cdot\|^* = \delta_{\|\cdot\|_* \leq 1}$$

where $\|\cdot\|_*$ is given by

$$\|v\|_* = \sup_x \{\langle v, x\rangle \mid \|x\| \leq 1\}$$

In particular, the *Moreau Decomposition Theorem* gives that

$$\mathbf{prox}_{\|\cdot\|}(v) = v - \mathbf{P}_{\{x\,:\,\|x\|_* \leq 1\}}(v)$$

1.3 Derivations of the proximal maps

We provide a quick derivation of the proximal operators of the L1 norm, the L2 norm, the iso norm, and the box prox. By the Moreau decomposition, we naturally obtain the proximal operators of the Fenchel conjugates. Let $b \in \mathbb{R}^d$ be fixed.

- Let $g : y \mapsto t\|y - b\|_1$. We have

  $$\mathbf{prox}_g(y) := \arg\min_u \{\, t\|u-b\|_1 + \frac{1}{2}\|y-u\|_2^2 \,\} = \begin{cases} b & y - b \in [-t, t] \\ y - \mathrm{sign}(y-b)\cdot t & \text{otherwise} \end{cases}$$

  We consider first-order optimality conditions, with the following cases:

  - If $u - b > 0$, then

    $$t - y + u = 0 \quad \Leftrightarrow \quad u = y - t$$

  - If $u - b < 0$, then

    $$-t - y + u = 0 \quad \Leftrightarrow \quad u = y + t$$

  - If $u - b = 0$, then $u = b$ under which by optimality conditions we have

    $$0 \in t \cdot [-1, 1] - y + b \quad \Leftrightarrow \quad y - b \in [-t, t]$$

- Now consider $g : y \mapsto t\|y - b\|_2^2$. This g is continuously differentiable so we can take its gradient directly and set it to zero, obtaining:

  $$\mathbf{prox}_g(y) = \frac{y+b}{2t+1}$$

- Consider now $g : (y_1, y_2) \mapsto t\gamma\|(y_1, y_2)\|_{\mathrm{iso}}$ where

  $$\|(x,y)\|_{\mathrm{iso}} = \sum_{k=1}^n (x_k^2 + y_k^2)^{\frac{1}{2}}$$

  To compute the proximal operator of g, we use Moreau decomposition (7.5.8), the generalized sum rule (7.5.4), and some subdifferential calculus (7.5.2). The Fenchel conjugate of g is $g^*(y_1, y_2) = t\gamma \max_{1 \leq k \leq n}(y_{1k}^2 + y_{2k}^2)^{1/2}$ where $y_1, y_2 \in \mathbb{R}^n$ (simplified after elementary conjugacy operations). So we have

  $$\mathbf{prox}_g(y_1, y_2) = (y_1, y_2) - \mathbf{prox}_{g^*}(y_1, y_2)$$

where

$$\mathbf{prox}_{g^*}(y_1, y_2) = \underset{(u,v)}{\arg\min} \left\{ t\gamma \max_{1\le k\le n} (u_k^2 + v_k^2)^{1/2} + \frac{1}{2}\|(y_1, y_2) - (u, v)\|_2^2 \right\}$$

By the generalized sum rule for the first line,

$$\begin{aligned}
\partial\, \mathbf{prox}_{g^*}(y_1, y_2) &= \partial\left(t\gamma \max_{1\le k\le n}(u_k^2+v_k^2)^{1/2}\right) + \partial\left(\frac{1}{2}\|(y_1,y_2)-(u,v)\|_2^2\right) \\
&= \partial\left(t\gamma \max_{1\le k\le n}(u_k^2+v_k^2)^{1/2}\right) + \nabla\left(\frac{1}{2}\|(y_1,y_2)-(u,v)\|_2^2\right) \\
&= t\gamma \cdot \operatorname{conv}\left(\nabla((u,v)\mapsto (u_k^2+v_k^2)^{1/2})\right) + \nabla\left(\frac{1}{2}\|(y_1,y_2)-(u,v)\|_2^2\right) \quad 7.5.2 \\
&= \left(\frac{t\gamma\cdot y_{1k}}{(y_{1k}^2+y_{2k}^2)^{1/2}} - y_{1k} + u_k,\ \frac{t\gamma\cdot y_{2k}}{(y_{1k}^2+y_{2k}^2)^{1/2}} - y_{2k} + v_k\right) \quad \text{by def. of convex hull}
\end{aligned}$$

which vanishes for each k when

$$u = \left(1 - \frac{t\gamma}{(y_{1k}^2+y_{2k}^2)^{1/2}}\right) y_1 \qquad v = \left(1 - \frac{t\gamma}{(y_{1k}^2+y_{2k}^2)^{1/2}}\right) y_2$$

as long as $(y_{1k}^2+y_{2k}^2)^{1/2} > 0$. If $(y_{1k}^2+y_{2k}^2)^{1/2} = 0$, then $y_{1k} = y_{2k} = 0$ so $(u,v) = (0,0)$.

We remark that in the original paper cite, they define the proximal map of f at x to be $\arg\min_u f(u) - 1/2\|u-x\|^2$ hence the difference in signs, although the same.

- For the box prox, $f : x \mapsto t\delta_S(x)$ where $S := \{x : x \in [0,1]\}$, the proximal operator is the projection. That is,

$$\begin{aligned}
\mathbf{prox}_f(x) &= \underset{u}{\arg\min}\{t\delta_S(u) + \frac{1}{2}\|x-u\|_2^2\} \\
&= t\cdot \underset{u}{\arg\min}\{\delta_S(u) + \frac{1}{2t}\|x-u\|_2^2\} \\
&= \underset{u\in S}{\arg\min} \frac{1}{2}\|x-u\|_2^2
\end{aligned}$$

2 Algorithms

The algorithms we present are used to solve the non-blind image deblurring problem, which is a convex optimization problem of the form

$$\begin{gathered}\min_{x,y} f(x) + g(y) \\ \text{subject to } Ax = y,\end{gathered} \tag{7}$$

where $f, g \in \Gamma$ have inexpensive prox-operators. We are given a vectorized, blurred image b, along with the kernel K used to blur it using the model

$$Kx + \eta = b, \tag{8}$$

where x is the original image, and η the added noise. Our goal is to recover the original image x from the blurred and noisy image b. We formulate the problem in two ways, namely, one which uses the L^1 norm, and the other the L^2 norm. First,

$$\min_x \|Kx - b\|_1 + \delta_S(x) + \gamma\|Dx\|_{\text{iso}}, \tag{9}$$

and second,

$$\min_x \|Kx - b\|_2^2 + \delta_S(x) + \gamma\|Dx\|_{\text{iso}}, \tag{10}$$

where $S = \{x \mid 0 \le x \le 1\}$, so that δ_S aims to restrict the range of allowed pixels.

The algorithms are based on a splitting scheme to facilitate computation of the resolvents of the maximal monotone operators. As mentioned in a bit more detail in the project prompt, we have from the fixed point iteration scheme

$$x^k = (I + t\mathcal{F})^{-1}(x^{k-1}) \qquad k = 1,2,...$$

where $\mathcal{F}$ is a maximal monotone operator. Splitting algorithms decompose the problem into

$$\begin{aligned} x^k &= (I + t\mathcal{A})^{-1}(z^{k-1}) \\ y^k &= (I + t\mathcal{B})^{-1}(2x^k - z^{k-1}) \\ z^k &= z^{k-1}\rho(y^k - x^k) \end{aligned}$$

where $t > 0$ is the step size and $\rho \in (0,2)$ is a relaxation parameter.

2.1 Primal Douglas-Rachford splitting (Spingarn's method)

This first algorithm, which we will refer to by PDRS, was invented in the 1950s as a method to solve the heat equation [1]. Per (9), let

$$\begin{aligned} f(x) =& \delta_S(x) \\ g(x) =& \|Kx - b\|_1 + \gamma\|Dx\|_{\text{iso}} \end{aligned} \tag{11}$$

and consider the minimization problem

$$\min_{x,y} f(x) + g(y) + \delta_0\left(\begin{bmatrix} A & -I \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix}\right) \tag{12}$$

with the optimality condition

$$0 \in \partial f(x) + \partial g(y) + \partial \delta_0\left(\begin{bmatrix} A & -I \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix}\right) =: \mathcal{F}(x,y) \tag{13}$$

As stated in the literature [2], $\mathcal{F}$ is the subdifferential of closed, convex functions, hence is a maximal monotone operator. Thus, by (5), the zeros of $\mathcal{F}$ are the fixed points of the resolvent of $\mathcal{F}$. Therefore, if we can find the fixed points of $\mathcal{F}$, we'll have found the optimal points. Unfortunately, this particular problem is too computationally expensive, so the strategy is to split $\mathcal{F}$ into two maximal monotone operators $\mathcal{A}$ and $\mathcal{B}$ which have inexpensive proximal operators. Finding the fixed points of $\mathcal{A}+\mathcal{B}$ will be equivalent to finding that of $\mathcal{F}$. Define $\mathcal{A}$ and $\mathcal{B}$ as,

$$\begin{aligned} \mathcal{A}(x,y) &= \partial f(x) + \partial g(y) = \partial(\delta_S(x)) + \partial(\|Ky - b\|_1 + \gamma\|Dy\|_{\text{iso}}) \\ \mathcal{B}(x,y) &= \partial\delta_0\left(\begin{bmatrix} \begin{bmatrix} K \\ D \end{bmatrix} & -I \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix}\right) \end{aligned} \tag{14}$$

We now focus on finding the resolvents of $\mathcal{A}$ and $\mathcal{B}$,

$$\begin{aligned} J_{\mathcal{A}} = (I + \lambda\mathcal{A})^{-1} &= \mathbf{prox}_{\lambda(f+g)}(x,y) \quad \text{[by (4)]} \\ &= (\mathbf{prox}_{\lambda f}(x), \mathbf{prox}_{\lambda g}(y)) \quad \text{[by Prop. 3]} \end{aligned} \tag{15}$$

where

$$\mathbf{prox}_{\lambda f}(x) = \mathbf{prox}_{\lambda\delta_S}(x) = \arg\min_u \frac{1}{2\lambda}\|x-u\|_2^2 + \delta_S(x) = \arg\min_{u\in S} \frac{1}{2\lambda}\|x-u\|_2^2 = \mathbf{P}_S(x), \tag{16}$$

which is the euclidean projection operator onto S. It can be computed element-wise by

$$(\mathbf{P}_S(x))_i = \begin{cases} 0, & \text{if } x_i < 0 \\ x_i, & \text{if } x_i \in S \\ 1, & \text{if } x_i > 1 \end{cases} = \max\{0, \min\{x_i, 1\}\} \tag{17}$$

Now, we observe that $g(y)$ described in (11) can be written in a different form by letting

$$g(y_1, y_2, y_3) = \|y_1 - b\|_1 + \gamma\|(y_2, y_3)\|_{\text{iso}} \tag{18}$$

subject to

$$y_1 = Ky \quad \text{and} \quad \begin{bmatrix} y_2 \\ y_3 \end{bmatrix} = Dy \iff \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} K \\ D \end{bmatrix} y$$

Then, by Proposition 3, notice that the proximal operator of g may be obtained by splitting its components into

$$\varphi(y_1) = \|y_1 - b\|_1 \quad \text{and} \quad \psi(y_2, y_3) = \gamma\|(y_2, y_3)\|_{\text{iso}}$$

which gives

$$\mathbf{prox}_{\lambda g}(y_1, y_2, y_3) = (\mathbf{prox}_{\lambda\varphi}(y_1), \mathbf{prox}_{\lambda\psi}(y_2, y_3)) \tag{19}$$

We know that the proximal operator of the L^1 norm is the soft-thresholding operator T_λ, so $\mathbf{prox}_{\lambda\varphi}(y_1)$ is given element-wise as follows,

$$(\mathbf{prox}_{\lambda\varphi}(y_1))_i = (\mathbf{prox}_{\lambda\|\cdot\|_1}(y_1 - b))_i = T_\lambda((y_1 - b)_i) := \begin{cases} (y_1 - b)_i - \lambda, & \text{if } (y_1 - b)_i > \lambda \\ (y_1 - b)_i + \lambda, & \text{if } (y_1 - b)_i < -\lambda \\ 0, & \text{otherwise.} \end{cases} \tag{20}$$

As a result of Proposition 4, $\mathbf{prox}_{\lambda\psi}(y_2, y_3)$, is given element-wise by

$$(\mathbf{prox}_{\lambda\psi}(y_2, y_3))_i = \mathbf{prox}_{\gamma\lambda\|\cdot\|_{\text{iso}}}(y_2, y_3)_i = \alpha_i((y_2)_i, (y_3)_i),$$

where,

(provide a proof of this)

$$\alpha_i = \begin{cases} 1 - \frac{\gamma\lambda}{\sqrt{(y_2)_i^2 + (y_3)_i^2}}, & \text{if } \sqrt{(y_2)_i^2 + (y_3)_i^2} > \gamma\lambda \\ 0, & \text{otherwise.} \end{cases}$$

As for the resolvent of $\mathcal{B}$, it corresponds to the projection (x, y) of a vector $(\bar{x}, \bar{y})$ onto the subspace $\{(x, y) \mid [K \quad D]^T x = y\}$, expressed as

$$J_{\mathcal{B}} = (I + \lambda\mathcal{B})^{-1} = \left(x = \left(I + [K \quad D] \begin{bmatrix} K \\ D \end{bmatrix} \right)^{-1} \left(\bar{x} + \begin{bmatrix} K \\ D \end{bmatrix} \bar{y} \right), y = \begin{bmatrix} K \\ D \end{bmatrix} x \right)$$

We now present the algorithm:

Algorithm 1: Primal Douglas-Rachford Splitting

Input : Blurred image b, step size t, relaxation parameter ρ
Output : Deblurred image x
Initialize : (z_1^0, z_2^0)

for $k = 1, 2, \ldots,$ **do**
  $x^k \leftarrow \mathbf{prox}_{tf}(z_1^{k-1})$; `// Performs resolvent of` $\mathcal{A}$
  $y^k \leftarrow \mathbf{prox}_{tg}(z_2^{k-1})$;
  $u^k \leftarrow (I + A^T A)^{-1}\left(2x^k - z_1^{k-1} + A^T(2y^k - z_2^{k-1})\right)$; `// Performs resolvent of` $\mathcal{B}$
  $v^k \leftarrow A(u^k)$;
  $z_1^k \leftarrow z_1^{k-1} + \rho(u^k - x^k)$;
  $z_2^k \leftarrow z_2^{k-1} + \rho(v^k - y^k)$;
end
return $x\text{Sol} = \mathbf{prox}_{tf}(z^k)$;

2.2 Primal-Dual Douglas Rachford Splitting

In this algorithm, we consider the dual problem,

$$\text{maximize } -f^*(-A^T z) - g^*(z) \tag{21}$$

with the optimality conditions for both the primal and the dual problem

$$0 \in \begin{bmatrix} 0 & 0 & A^T & I \\ 0 & 0 & -I & 0 \\ -A & I & 0 & 0 \\ -I & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} + \begin{bmatrix} 0 \\ \partial g(y) \\ 0 \\ \partial f^*(w) \end{bmatrix}$$

In the above, w is the dual variable and z is the dual multiplier for the constraint $y = Ax$, thus the dual problem is subject to

$$A^T z + w = 0$$

We reduce this to a smaller system by using the relationship $\partial g^* = (\partial g)^{-1}$, which follows from the fact that g is closed and convex, and by the properties of the subgradient and conjugate. We can eliminate y by using the above equality that $\partial g^* = (\partial g)^{-1}$, which would leave us with

$$0 \in \begin{bmatrix} 0 & A^T & I \\ -A & 0 & 0 \\ -I & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ z \\ w \end{bmatrix} + \begin{bmatrix} 0 \\ \partial g^*(z) \\ \partial f^*(w) \end{bmatrix}$$

We can further reduce this by eliminating w, using a similar equality that $\partial f = (\partial f^*)^{-1}$, we get

$$0 \in \begin{bmatrix} 0 & A^T \\ -A & 0 \end{bmatrix} \begin{bmatrix} x \\ z \end{bmatrix} + \begin{bmatrix} \partial f(x) \\ \partial g^*(z) \end{bmatrix} \tag{22}$$

We can now split the monotone operator we have above to a sum of two monotone operators given as

$$\mathcal{A}(x,z) = \begin{bmatrix} \partial f(x) \\ \partial g^*(x) \end{bmatrix} \quad \mathcal{B}(x,z) = \begin{bmatrix} 0 & A^T \\ -A & 0 \end{bmatrix} \begin{bmatrix} x \\ z \end{bmatrix} \tag{23}$$

Similar to the previous algorithm, we now find the resolvents of $\mathcal{A}$ and $\mathcal{B}$.

As seen before with the resolvent of $\mathcal{A}$, the value $(x,y) = (I + At)^{-1}(\hat{x},\hat{z})$ is given by

$$x = \mathbf{prox}_{tf}(\hat{x}) \quad z = \mathbf{prox}_{tg^*}(\hat{z})$$

For z, we can apply moreau decomposition to get an alternate expression dealing with the conjugate g^*, thus we have,

$$z = \hat{z} - t\,\mathbf{prox}_{t^{-1}g}(\hat{z}/t)$$

Simplifying the above, we are left with,

$$z = \hat{z} - \mathbf{prox}_{gt}(\hat{z})$$

This will be a similar computation to the prox operator in algorithm 1, but we have right scalar multiplication in this case.

The resolvent for $\mathcal{B}$ is a linear mapping, so considering resolvents for skew-symmetric linear operators

$$\begin{aligned}(I + t\mathcal{B})^{-1} &= \begin{bmatrix} I & tA^T \\ -tA & I \end{bmatrix}^{-1} \\ &= \frac{1}{I - ((tA^T)(-tA))} \begin{bmatrix} I & -tA^T \\ tA & I \end{bmatrix} \\ &= \frac{1}{I + t^2 A^T A} \begin{bmatrix} I & -tA^T \\ tA & I \end{bmatrix}\end{aligned}$$

$$= \begin{bmatrix} 0 & 0 \\ 0 & I \end{bmatrix} + \frac{1}{I+t^2A^TA} \begin{bmatrix} I \\ tA \end{bmatrix} \begin{bmatrix} I & -tA^T \end{bmatrix}$$
$$= \begin{bmatrix} 0 & 0 \\ 0 & I \end{bmatrix} + \begin{bmatrix} I \\ tA \end{bmatrix} (I+t^2A^TA)^{-1} \begin{bmatrix} I \\ -tA \end{bmatrix}^T$$

Letting (w,v) be the value of the resolvent of $\mathcal{B}$,

$$\begin{aligned}
\begin{bmatrix} w \\ v \end{bmatrix} &= \begin{bmatrix} I & tA^T \\ -tA & I \end{bmatrix}^{-1} \begin{bmatrix} \hat{x} \\ \hat{z} \end{bmatrix} \\
&= \left(\begin{bmatrix} 0 & 0 \\ 0 & I \end{bmatrix} + \begin{bmatrix} I \\ tA \end{bmatrix} (I+t^2A^TA)^{-1} \begin{bmatrix} I \\ -tA \end{bmatrix}^T \right) \begin{bmatrix} \hat{x} \\ \hat{z} \end{bmatrix} \\
&= \begin{bmatrix} 0 & 0 \\ 0 & I \end{bmatrix} \begin{bmatrix} \hat{x} \\ \hat{z} \end{bmatrix} + \begin{bmatrix} I \\ tA \end{bmatrix} (I+t^2A^TA)^{-1} \begin{bmatrix} I \\ -tA \end{bmatrix}^T \begin{bmatrix} \hat{x} \\ \hat{z} \end{bmatrix} \\
&= \begin{bmatrix} 0 \\ \hat{z} \end{bmatrix} + \begin{bmatrix} I \\ tA \end{bmatrix} (I+t^2A^TA)^{-1} \begin{bmatrix} I & -tA^T \end{bmatrix} \begin{bmatrix} \hat{x} \\ \hat{z} \end{bmatrix} \\
&= \begin{bmatrix} 0 \\ \hat{z} \end{bmatrix} + \begin{bmatrix} I \\ tA \end{bmatrix} (I+t^2A^TA)^{-1} (\hat{x} - tA^T\hat{z}) \\
&= \begin{bmatrix} 0 \\ \hat{z} \end{bmatrix} + \begin{bmatrix} I \\ tA \end{bmatrix} \frac{\hat{x} - tA^T\hat{z}}{(I+t^2A^TA)} \\
&= \begin{bmatrix} 0 \\ \hat{z} \end{bmatrix} + \begin{bmatrix} \frac{\hat{x} - tA^T\hat{z}}{(I+t^2A^TA)} \\ tA\left(\frac{\hat{x} - tA^T\hat{z}}{(I+t^2A^TA)}\right) \end{bmatrix} \\
&= \begin{bmatrix} \frac{\hat{x} - tA^T\hat{z}}{(I+t^2A^TA)} \\ \hat{z} + tA\left(\frac{\hat{x} - tA^T\hat{z}}{(I+t^2A^TA)}\right) \end{bmatrix}
\end{aligned}$$

Thus we have the solution for both w, v,

$$w = \frac{\hat{x} - tA^T\hat{z}}{(I+t^2A^TA)} = \frac{\hat{x}}{(I+t^2A^TA)} - \frac{tA^T\hat{z}}{(I+t^2A^TA)} \tag{24}$$

$$v = \hat{z} + tA\left(\frac{\hat{x} - tA^T\hat{z}}{(I+t^2A^TA)}\right) = \hat{z} + \frac{t\hat{A}x}{(I+t^2A^TA)} - \frac{t^2AA^T\hat{z}}{(I+t^2A^TA)} \tag{25}$$

Notice in the algorithm, we will let $\hat{x} = 2x^k - p^{k-1}$ and $\hat{z} = 2z^k - p^{k-1}$.

Here is the Primal-Dual Douglas Rachford splitting algorithm,

Algorithm 2: Primal-Dual Douglas-Rachford Splitting

Input : Blurred image b, step size t, relaxation parameter ρ
Output : Deblurred image x
Initialize : p^0, q^0

for $k = 1, 2, \ldots,$ **do**

- $x^k \leftarrow \mathbf{prox}_{tf}(p_1^{k-1})$; // Performs resolvent of $\mathcal{A}$
- $z^k \leftarrow \mathbf{prox}_{tg^*}(q^{k-1})$;
- $\begin{bmatrix} w^k \\ v^k \end{bmatrix} = \begin{bmatrix} I & tA^T \\ -tA & I \end{bmatrix}^{-1} \begin{bmatrix} 2x^k - p^{k-1} \\ 2z^k - q^{k-1} \end{bmatrix}$; // Performs resolvent of $\mathcal{B}$
- $p^k \leftarrow p^{k-1} + \rho(w^k - x^k)$;
- $q^k \leftarrow q^{k-1} + \rho(v^k - z^k)$;

end

return $x\mathrm{Sol} = \mathbf{prox}_{tf}(p^k)$;

2.3 ADMM (Dual Douglas-Rachford)

The dual problem from above

$$\text{maximize}_y \ -f^*(-A^T y) - g^*(y) \tag{26}$$

has the equivalent form

$$\text{minimize}_z \ -f^*(-A^T z) - g^*(z) := h(z) + k(z)$$

and it is said that applying the Douglas-Rachford algorithm to this dual problem results in the alternating direction method of multipliers (ADMM) algorithm. That is, ADMM is uses the Douglas-Rachford iteration on the dual problem. Optimality is guaranteed when $0 \in \partial h(z) + \partial k(z)$ in which we have two maximal monotone operators with resolvents $\mathbf{prox}_{th}(z)$ and $\mathbf{prox}_{tk}(z)$.

ADMM like the previous algorithms, is a splitting algorithm. Unlike the previous two algorithms, it has different optimality conditions, that is it uses solely optimality conditions of the dual problem.

A form of the Douglas-Rachford iteration on the dual can be derived to be the following, as given in the project prompt:

$$\begin{aligned}
\hat{x}^k &= \arg\min_x \left(f(x) + (z^{k-1} - w^{k-1})^T Ax + \frac{t}{2}\|Ax\|^2 \right) \\
v^k &= z^{k-1} - w^{k-1} + tA\hat{x}^k \\
\hat{y}^k &= \arg\min_y \left(g(y) - (v^k + w^{k-1})^T y + \frac{t}{2}\|y\|^2 \right) \\
z^k &= v^k + w^{k-1} - t\hat{y}^k \\
w^k &= w^{k-1} + \rho v^k + (1-\rho) z^{k-1} - z^k.
\end{aligned}$$

In particular, we are interested in the general problem

$$\min_{u,y} f(u) + g(y)$$

$$\text{subject to } \begin{bmatrix} I \\ A \end{bmatrix} x + \begin{bmatrix} -I & 0 \\ 0 & -I \end{bmatrix} \begin{bmatrix} u \\ y \end{bmatrix} = 0$$

for which we have an augmented Lagrangian given as

$$L(x,u,y,w,z) = f(u) + g(y) + w^T(x-u) + z^T(Ax-y) + \frac{t}{2}\left(\|x-u\|^2 + \|Ax-y\|^2\right)$$

which makes no difference in the above formulation as we alternate minimization over x and minimization over (u,y), so that they are constants in alternation.

Algorithm 3: Alternating Direction Method of Multipliers (ADMM)

Input : Blurred image b, step size t, relaxation parameter ρ
Output : Deblurred image x
Initialize : x^0, u^0, y^0, w^0, z^0

for $k = 1, 2, \ldots,$ **do**
  $x^k \leftarrow (I + A^T A)^{-1}\left(u^{k-1} + A^T y^{k-1} - \frac{1}{t}(w^{k-1} + A^T z^{k-1})\right)$;
  $u^k \leftarrow \text{prox}_{t^{-1}f}(\rho x^k + (1-\rho)u^{k-1} + w^{k-1}/t)$;
  $y^k \leftarrow \text{prox}_{t^{-1}g}(\rho A x^k + (1-\rho)y^{k-1} + z^{k-1}/t)$;
  $w^k \leftarrow w^{k-1} + t(x^k - u^k)$;
  $z^k \leftarrow z^{k-1} + t(Ax^k - y^k)$;
end
return $x\text{Sol} = (I + A^T A)^{-1}\left(u^k + A^T y^k - \frac{1}{t}(w^k + A^T z^k)\right)$;

2.4 Chambolle-Pock

The Chambolle-Pock method considers the generic optimization problem

$$\begin{aligned} &\min_{x,y} f(x) + g(y) \\ &\text{subject to } Ax = y, \end{aligned} \tag{27}$$

by finding the solution of

$$\min_x \max_y \langle Ax, y\rangle + f(x) - g^*(y). \tag{28}$$

It is mentioned that in our case, the solution turns out to be the Lagrangian of our generic optimization problem. decription of the similarities and differences of CHambolle Pock from the other algos The algorithm proceeds as below.

Algorithm 4: Chambolle-Pock Algorithm

Input : Blurred image b, step sizes t, s
Output : Deblurred image x
Initialize : x^0, y^0, z^0

for $k = 1, 2, \ldots,$ **do**
  $y^k = \text{prox}_{sg^*}(y^{k-1} + sAz^{k-1})$;
  $x^k = \text{prox}_{tf}(x^{k-1} - tA^T y^k)$;
  $z^k = 2x^k - x^{k-1}$;
end
return $x\text{Sol} = x^k$;

summarize the following (or remove)

By choosing the step sizes s and t such that $st(\|A\|_2^2) < 1$, we can ensure the following convergence:

$$\frac{\|y^k - \hat{y}\|_2^2}{2s} + \frac{\|x^k - \hat{x}\|_2^2}{2t} \leq C\left(\frac{\|y^0 - \hat{y}\|_2^2}{2s} + \frac{\|x^0 - \hat{x}\|_2^2}{2t}\right). \tag{29}$$

This establishes that the sequence of iterates, (x^k, y^k), is bounded. Therefore one can find a subsequence (x^{k_m}, y^{k_m}) that converges to some limit (x^*, y^*). It can also be shown that x^{k_m-1} and y^{k_m-1} also converge respectively to x^* and y^*. This shows that (x^*, y^*), obtained as a limit of the above algorithm, is a fixed point of the iterations, and hence a saddle point of the min-max problem (28).

There are a couple of important differences between the Chambolle-Pock algorithm and the previous variants of the Douglas-Rachford algorithm. One, the step sizes s and t in Chambolle-Pock are dependent on the size of the operators in the problem (i.e. the maximum singular value of matrix A). Second, this algorithm does not require solving a system of linear equations involving $I + A^T A$. We only need vector-matrix multiplications of A and A^T.

Despite these operational differences, at its core, the Chambolle-Pock algorithm can be interpreted as a form of preconditioned ADMM. The first step of ADMM solves a least squares problem, which involves a difficult matrix inversion. In general, this could be quite an expensive operation. To get around this, the Chambolle-Pock algorithm adds

$$\frac{1}{2}\langle \left(\frac{1}{s} - tAA^T\right)(y - y^{k-1}), y - y^{k-1}\rangle$$

to the augmented Lagrangian to be minimized at the k-th iteration:

$$y^k = \arg\min_y g^*(y) - \langle A^T y, x^{k-1}\rangle + \frac{t}{2}\|A^T y + z^{k-1}\| + \frac{1}{2}\langle(\frac{1}{s} - tAA^T)(y - y^{k-1}), y - y^{k-1}\rangle$$

This simplifies to

$$y^k = (I + s\partial g^*)^{-1}(y^{k-1} + sAz^{k-1})$$

where

$$z^k = 2x^k - x^{k-1},$$

which are exactly two of the steps of the algorithm described above.

3 Tuning and Results

it is said to enter the best parameters as the default

Table 1: Default Parameters Used

Hyperparameter	**In-line Name** `i.`	**Value**
Kernel	`kernel`	
Noise		
Total Iteration	`maxiter`	500
gamma L1	`gammal1`	0.1
gamma L2	`gammal2`	0.02
t	`tprimaldr`	
	`rhoprimaldr`	
	`tprimaldualdr`	
	`rhoprimaldualdr`	
	`tadmm`	
	`rhoadmm`	
	`scp`	
	`tcp`	

3.1 Hyperparameter tuning

Kernel and noise

Gamma

Step size and relaxation parameter

3.2 Results

Acknowledgments

Please include a list of each person (with student ID) in your group and write a paragraph **for each person** with exactly what they did. Note that the Acknowledgment section does not count towards the page limit.

Example Acknowledgments would look like:

Alexandre St-Aubin (Student ID: 261115785) We ball.

Jonathan Campana (Student ID: 260985613) We ball.

Elliot Forcier-Poirier (Student ID:) We ball.

Edmund Wu (Student ID: 261033501) We ball.

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A Appendix

Optionally include extra information (complete proofs, additional experiments and plots) in the appendix. This section will often be part of the supplemental material.